

Megh Bahadur KC

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Google Scholar: <https://scholar.google.com/citations?user=-qMDe58AAAAJ&hl=en&authuser=1>

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PROFESSIONAL SUMMARY

Transportation Engineer at California Department of Transportation & Ph.D. candidate in Civil Engineering at the University at Buffalo, SUNY, specializing in sustainable public transport, freight systems and ITS through optimization, machine learning and deep reinforcement learning. Over 10 years of combined research, academic and industry experience developing efficient optimization tools, predictive models, techno-economic analyses, advanced data analytics and inclusive planning/policies at National Laboratory of the Rockies (NRL), New York City DOT and Utah State University. Proven achievements include 7+ peer-reviewed publications, 17+ conference presentations, 2 internships, multiple awards and fellowships. Effective undergraduate and graduate level teaching and hands-on project management as a Project Manager.

EDUCATION

Ph.D. in Civil Engineering; **University at Buffalo, SUNY**- GPA: 4.00, by **Nov 2026**

Dissertation Title: “**Accelerating Bus Transit Electrification through Optimization, Metaheuristics Algorithm and Reinforcement Learning Frameworks**” Advisor: Dr. Ziqi Song, Associate Professor

M. Sc. in Construction Management; Pokhara University, Nepal- GPA: 3.95

B. E. in Civil Engineering, Institute of Engineering, Tribhuvan University, Nepal; ranked top 4

RESEARCH INTERESTS

- **Operations Research:** Optimization, Application of OR on Electrified Transportation Systems, Reinforcement Learning for Transport System Efficiency, Multimodal Freight System Modeling, Supply Chain Management
- AI/RL/ML in Transportation, ITS and Advanced Transportation Applications (V2X, CAV), Digital Twins
- Agent-Based Modeling, Simulation, Travel Demand Modeling, Discrete Choice Modeling, Geospatial Analysis

WORK EXPERIENCE

SN.	From	To	Employer	Position	Supervisor
8	07-2026	To date	California Department of Transportation	Transportation Engineer	Nandini Vishwanath
7	01- 2025	to date	University at Buffalo, SUNY, NY	Graduate Research Assistant (GRA)	Asso. Prof. Dr. Ziqi Song
6	09-2025	09-2026	New York City Department of Transportation, New York, NY	Freight Policy Tool Development Intern	Catherine Ponte, Team Lead
5	05-2024	01-2025	National Laboratory of the Rockies, Golden, CO	Supply Chain Modeling Intern	Jason Lustbador, Group Manager
4	08- 2021	01-2025	Utah State University, Logan, UT	GRA	Asso. Prof. Dr. Ziqi Song
3	03- 2020	08- 2021	Pokhara University, Nepal	Assistant Professor	Dr. KN Pandey, Principal
2	07- 2015	08- 2021	Ministry of Physical Infrastructure and Transport, Nepal	Highway Engineer	Director General, Department of Roads
1	03- 2014	02- 2015	Tribhuvan University, Nepal	Assistant Lecturer	Prof. Dr. Rajan Suwal

PROFESSIONAL AFFILIATIONS

- **Editor/Paper Review Coordinator**, Transportation Research Board (TRB/TRR), 2026-2027
- Member, TRB Standing Technical Committee on Transit Operations (AP014), 08/2025-04/2028
- Member, TRB Standing Committee on Bus Transit Systems (AP050), 2025
- Member, TRB Sub-committee on Freight Modeling (AT015(1)), 2023-2025
- Member, American Society of Civil Engineers (ASCE), 2025-2027

1) Graduate Research Assistant @ University at Buffalo, SUNY

- PhD Dissertation research:
 - Multi-Agent Deep Reinforcement Learning for Integrated Bus Scheduling and Charging Management Under Uncertainty in Multi-Line Public Transit Operation. *[to be submitted to TR-Part C]*
 - Deep Reinforcement Learning Framework for Simultaneous Fleet Optimization and Charging for Multi-line Bus Transit System in a Dynamic Environment. *[under review, TR-Part C]*
 - Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Bus System Including Demand Charges & Battery Degradation. *[under review, Applied Energy]*
 - Bus Electrification: Fleet Optimization and Charging Management Considering Partial Recharges, TOU and Demand Charge. *[under review, Transportation Research Interdisciplinary Perspectives]*
- Leveraging data pipelines, HPC and scalable architectures for complex computational models and big data
- Developed and published *freight-analytics-dashboard* Python package on PyPI for reusable data workflows
- Led "Residential Location Choice Modeling" project for WFRM Utah, identified key policy interventions for TOD & pedestrian-friendly infrastructure, contributing to an increase in transit ridership

GRANT/PRINCIPAL INVESTIGATOR RESEARCH

Role	Award #	Federal Sponsor	Sponsor	Award Title	Project Title	Amount	Award Date
Subawardee PI	PN23(07)8 Subaward #4024	USDOT-FHWA	New York Metropolitan Transportation Council/New York State Department of Transportation	9/11th Memorial Program for Regional Transportation Planning	Freight VMT and Emissions Assessment Tool	USD 35,000	10-2025

Proposal Writing Contribution

- NYSERDA RFP 5965- *Electric School Bus Transition Support Contractor*, \$1M, 2025-2028
 - Drafted Statement of Work (SoW) section
- DOE, Joint Office of Energy and Transportation (JOET), Concept paper in response to FOA – *Expanding E-Mobility Solutions through Electrified School Buses*, \$250,000.
 - Drafted Background and Proposed Methodology section

2) Transportation Engineer @ California Department of Transportation

- Sub-regional travel demand models and forecasts in support of traffic operational analyses and environmental studies for project alternatives studies.

3) Subawardee PI/Research Fellow @ New York City Department of Transportation (NYCDOT)

- Developed 4-step freight demand modeling and calibrated using real-time vehicle count dataset
- Developed a VMT assessment and GHG reduction tool for NYC freight transportation
- Quantified the effect of freight policy interventions such as NY congestion pricing, off-hour delivery, and clean truck programs in terms of VMT, GHGs, and air pollution
- Complex public and private big data analysis including StreetLight, INRIX, Geotab, ATRI

4) Supply Chain Modeling Intern @ National Laboratory of the Rockies (NRL)

- Developed & coded Freight Analysis Framework (FAF) & CFS flow containerization model & calibration
- Robust optimization of intermodal freight operations in disruptive events (national scale) through modeling & simulation of containerized freight flow
- Built last-mile delivery vehicle routing: VRP, CVRP, VRPPD, and benchmarked with Google OR-Tools
- Contributed on SMART 2.0 Freight project by successfully developing & integrating the electric variant of NREL's open-sourced Freight Integrated Simulation Model (FRISM) for heterogeneous mixed fleet combination using VNS metaheuristics
- Formulated methodological framework for the county-level freight flow disaggregation model algorithm

- Created & hosted an interactive Container Freight data visualization Dashboard in Streamlit

5) Graduate Research Assistant @ Utah State University

- Automated a space-time prism (STP) modeling by developing an ArcGIS Pro algorithm in Python using the ArcPy environment, achieving 95% process efficiency to measure accessibility for people with disabilities
- Developed and linearized a multi-objective optimization model in GAMS and produced a hybrid metaheuristic algorithm in Python 3.10 with solver efficiency (730 seconds) in large-scale networks
- Produced and enhanced a multistage multi-objective linear /nonlinear optimization model with the achievement of efficient recharging and scheduling of EVs in 1,000+ node networks in 2 stages
- Performed techno-economic analysis on “Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Electric School Bus System Including Demand Charges & Battery Ageing.”
- Conducted emission modeling for a new electrified metro, estimating mode shift, reduced 6% annual GHG
- Led the best group project in machine learning titled “Fake News Detection: Investigating ChatGPT” by using Naive Bayes, Random Forest, and Neural Networks among 19 groups
- Conducted research on Discrete Mode Choice Modeling using Nested and Multinomial Logit in Pandas Biogeme, resulting in policy recommendations for the Inclusive Transportation System in Utah
- Completed term project on Change Detection Analysis for Selected Municipalities in Utah using ArcGIS Pro
- Participated in Kaggle Competition as a part of Final Project on Machine Learning and Data Analytics

6) Transportation Engineer/ Project Manager @ Ministry of Transport Infrastructure

- Completed feasibility studies for multi-million-dollar Highway Projects, cost-benefit analysis, specifications.
- Obtained design approval for a 17-mile 6-lane road project, representing the Government of Nepal.
- Managed the procurement process for a massive ~\$40M road project, including bid document preparation, contract work specification, bid publication, evaluation, and contract award.
- Served as a site project manager, leading a team of 3 engineers, 8 sub-engineers, and 15 site supervisors.
- Skillfully coordinated with the communication manager for the upper body during the project's execution.
- Recognized with the Best Employee Award in 2019 for reducing office annual costs by 19% through optimizing employee schedules and operations.

RESEARCH EXPERIENCE

Peer-Reviewed Journal Conference Articles

- **KC, M. B., & Song, Z.** (2025). Evolving School Transportation: Integrated Dynamic Route Optimization and Partial Charging for Mixed Fleets. In arXiv e-prints. DOI: <https://doi.org/10.48550/arXiv.2511.00600>
- **KC, M. B., & Song, Z.** (2025). School Transport Electrification - Adoption, Strategies, Methods and Policy: A Comprehensive Review. In arXiv e-prints. DOI: <https://doi.org/10.48550/arXiv.2601.00838>
- **KC, M. B.** (2025). Land Use Transition and City Desirability: Case Study of Cache County Over Two Decades. *Growth and Change, A Journal of Urban and Regional Policy*, 2025; 56(4). DOI: <https://doi.org/10.1111/grow.70057>
- **KC, M.B., Song, Z., Park, K. & Christensen, K.** (2025). *Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah*. In arXiv e-prints. DOI: <https://doi.org/10.48550/arXiv.2511.11748>
- **KC, M.B., Mishra, A.K. & Karki, R.** (2020). Perception Based Assessment of Bidding Practices and Effectiveness of E-Bidding System in Nepal. *East African Scholars of Economics, Business and Management* 2020; 3(7): 588-597. [Link](#)
- **Mishra, A.K., KC, M. B. & Aithal, P.S.** (2020). Association of Number of Bidders and Minimum Bid Ratio (AER) with Effect of E-bidding of Different Project. *International Journal of Management, Technology, and Social Sciences* 2020; 5(2): 201-215. [Link](#)
- **KC, M. B., & Mishra K.** (2019). Bidding Trends of Contracts based on Types and Sizes of Projects under Road Divisions Butwal and Shivapur. *Journal of Advanced Research in Construction and Urban Architecture* 2019; 4(4): 7-16. [Link](#)

Under Review and Under Preparation Journal Papers

- **KC, M. B., & Song, Z.** (2026). Multi-Agent Deep Reinforcement Learning for Integrated Bus Scheduling and Charging Management Under Uncertainty in Multi-Line Public Transit Operation. [to be submitted to TR C].

- **KC, M. B.,** Song, Z. & Chaoyue Zhao. (2026). Deep Reinforcement Learning Framework for Integrated Fleet and Charging Management for Electric Bus Transit System in Dynamic Environment. [TRBAM 2027].
- **KC, M. B.,** & Song, Z. (2026). Bus Electrification: Fleet Optimization and Charging Management Considering Partial Recharges, TOU and Demand Charge. [TRBAM 2027, Under review TRIP].
- **KC, M. B.,** & Song, Z. (2025). Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Electric School Bus System Including Demand Charges & Battery Ageing. [TRB 2024, under review, Applied Energy].
- **KC, M. B.,** & Song, Z. (2025). Multi-Stage, Multi-School Electric Bus Routing and Scheduling Optimization with recharging schedule and re-use of Buses using metaheuristics. [TRB 2024].
- **KC, M. B.,** Song, Z., Park, K., Chamberlin, B. & Christensen, K. (2025). Household Residential Location Choice Modeling for People with Disabilities in Utah. [TRBAM 2026 & under review, TR Part A].

RESEARCH PROJECTS

- INtermodal Freight Optimization for a Resilient Mobility Energy System (**INFORMES**), Advanced Research Projects Agency-Energy's (ARPA-E), US **Department of Energy (DOE)** @ National Renewal Energy Laboratory, Denver, Co, USA
- Route Optimization, On-route Charging and V2G Integration on Transportation thrust for **ASPIRE** (Advancing Sustainability through Powered Infrastructure for Roadway Electrification) National Science Foundation (**NSF**) **Engineering Research Center**, Logan, Utah
- Increasing Affordability, Energy Efficiency, and Ridership of Transit Bus Systems through Large-Scale Electrification, US Department of Energy (DOE)
- Systems and Modeling for Accelerated Research in Transportation **SMART 2.0, Vehicle Technologies Office**, US Department of Energy (DOE) @ National Renewal Energy Laboratory, Denver CO
- Feasibility Analysis of Electric Roadways Through Localized Traffic, Cost, Adoption, and Environmental Impact Modeling, **ARPA-E, DOE**
- Investigating the Feasibility of Introducing Alternative Fuel Vehicles into Maintenance Fleet, Utah Department of Transportation (**UDOT**)
- Enhancement in 4-step Travel Demand Modeling considering individuals with disabilities incorporating physical community and environmental context, Funded by National Institute on Disability, Independent Living, and Rehabilitation Research (**NiDILRR**)

AWARDS

- **Winner – Fellowship (\$35,000)**, September 11th Memorial Program for Regional Transportation Planning; New York Metropolitan Transportation Council (NYMTC), 2025
- **Winner** - Institute of Transportation Engineers (ITE) Student Research Competition, Utah, 2024
- **Winner** - Prakash S Lad Scholarship for Courage and Perseverance in the College of Engineering (ENGR), Utah State University, 2023 (**\$2000.00**)
- Best student term project on data science titled "*Fake News Detection: Investigating ChatGPT.*"
- **Dean's List and University topper** in MSc Construction Management, Pokhara University, 2021
- **Outstanding MSc Student of the Year**, Pokhara University, 2019.

MEDIA MENTIONS

- Weitekamp, R. (2025, Oct). *Scholars Announced for NYMTC's 2025-26 9-11 Memorial Fellowship Program.* <https://www.nymtc.org/news-and-events#2025scholars>
- Maxwell, A. (2025, Sept 17). *CSEE student awarded September 11th Memorial fellowship for AI driven transportation solution for system efficiency.* <https://www.buffalo.edu/istl/news-events/news.host.html/content/shared/www/istl/istl-news/csee-student-awarded-september-11th-memorial-fellowship-for-ai-d.detail.html>
- Dahle, S. (2024, Jan 23). *USU Engineering Scholar Showcases Transportation Research.* <https://engineering.usu.edu/news/main-feed/2024/usu-engineering-scholar-showcases-transportation-research>

TEACHING EXPERIENCE

- **Teaching Assistant**, University at Buffalo; Utah State University, USA: 2021- present

- Taught, provided feedback & graded students on assignments in class and in office hours
- **Assistant professor**, Pokhara University, Nepal: 2020-2021
 - Project planning & Control, Graduate Course – achieved 100% pass rate for MSc 1st semester
- **Lecturer**, Tribhuvan University, Nepal: 2013-2014
 - Transportation Engineering, Undergraduate Course – improved pass rates than in previous year
 - Supervised 2 Undergraduate Final projects on transportation planning and design

SKILLS

- **Programming and Optimization:** Python, R, GAMS, GurobiPy, CPLEX
- **Mapping and Visualization Software:** ArcGIS Pro, QGIS, ArcPy, matplotlib, seaborn
- **Machine/Deep Learning Frameworks:** SVM, SVD, DQN, Pytorch, HPC, PPO, MAPPO, MADDPG
- **Transportation Demand Modeling:** CUBE, Synchro, TransCAD, VISSIM, SmartRoad
- **Project management software:** Microsoft Project
- **Others:** Jupyter, Pandas, NumPy, scikit-learn, API web scraping, NLP, SQL

CERTIFICATIONS & LICENCES

- California Engineer In Training (EIT) #188398
- General Engineer, Civil, Nepal Engineering Council (NEC), since 2015

PEER-REVIEWED CONFERENCE PRESENTATION

21. **KC, M. B.**, Song, Z. & Zhao, Chaoyue (2026). Online Deep Reinforcement Learning for Integrated Multi-line Electric Bus Scheduling and Charging under Operational Uncertainty. *Transportation Research Board (TRB) Annual Meeting, January 10-14, 2027*, Washington DC [accepted]
20. **KC, M. B.** & Ponte, Catherine (2026). Bridging Gap in Urban Freight Governance: A Strategic Freight Policy Screening Model for New York City. *Transportation Research Board (TRB) Annual Meeting, January 10-14, 2027*, Washington DC [accepted]
19. **KC, M. B.**, Song, Z., Park, K., Chamberlin, B. & Christensen, K. (2025). Household Residential Location Choice for People with Disabilities: Determinants & Implications for Inclusive Planning. *Transportation Research Board (TRB) Annual Meeting, January 5-9, 2025*, Washington DC
18. **KC, M. B.**, & Song, Z. (2025). Land Use Transition and City Desirability: Case Study of Cache County over Two Decades. *Transportation Research Board (TRB) Annual Meeting, January 11-15, 2025*, Washington DC
17. **KC, M. B.**, Song, Z., Park, K., Chamberlin, B. & Christensen, K. (2025). Household Residential Location Choice Modeling for People with Disabilities in Utah. *ASCE International Conference on Transportation Development*, June 6-8, 2025, Arizona, USA (Accepted)
16. **KC, M. B.**, & Song, Z. (2024). Multi-Stage, Multi-School Electric Bus Routing and Scheduling Optimization Using Metaheuristics. *Transportation Research Board (TRB) Annual Meeting, January 5-9, 2025*, Washington DC
15. **KC, M. B.**, & Song, Z. (2024). Integrated Optimized Charging and Economic Analysis of V2G Enabled School Bus System. *Transportation Research Board (TRB) Annual Meeting, January 5-9, 2025*, Washington DC
14. **KC, Megh** and Song, Ziqi (2024), " Inclusive Transport Policy: Mode Choice Behavior of People with Disabilities in Utah," *2024 ITE Mountain District Annual Meeting, Big Sky, Montana*, 19-21 June 2024 (Accepted)
13. **KC, Megh**; Song, Ziqi; Park, Keunhyun and Keith Christinsen (2024), "Inclusive Transportation System and Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah", *6-Minute Showcase for the 2024 TRB Annual Meeting as part of the Young Member Council - Sustainability and Resilience*, Washington DC (awarded as recognized presentation)
12. **KC, Megh** and Song, Ziqi (2024), "Evolving School Transportation: A Comprehensive Approach to Bus Electrification with Dynamic Route Optimization and Partial Charging for Mixed Fleets", *Transportation Research Board (TRB) Annual Meeting, January 7-11, 2024*, Washington DC

11. KC, Megh; Song, Ziqi; Park, Keunhyun and Keith Christinsen (2024), “Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah”, *Transportation Research Board (TRB) Annual Meeting*, January 7-11, 2024, Washington DC

RESEARCH OUTREACH ACTIVITIES

ROLE: REVIEWER

11. Transportation Research Board Annual meeting, 2026
10. International Journal of Transportation Engineering and Technology (IJTET)
9. Journal of Transportation System and Engineering (JoTSE)
8. Energy 360
7. Transportation Research Record, Journal of Transportation Research Board
6. Transportation Research Board Annual meeting, 2025
5. Transportation Research Board Annual meeting, 2024
4. Transportation Research Board Annual meeting, 2023

ROLE: EDITOR / PAPER REVIEW COORDINATOR

Article Title

30. Wheelchair Accessibility in Urban Multimodal Transit: Low-Floor Bus Allocation
29. Real-Time Coordinated Metro-Bus Rescheduling under Uncertain Metro Disruption via Multi-Agent Deep Reinforcement Learning
28. Identifying Edge Cases in Transit Bus Crashes Involving Pedestrians and Bicyclists through Hierarchical Clustering of Crash Narratives
27. Urban Transit Resilience Under Joint Demand and Supply Disruptions: A Multi-Scale Analysis of the New York City Subway
26. A mixed-integer linear formulation for the vehicle scheduling of electric modular buses
25. A Standardized Feasibility Assessment Framework for Planning Electric Bus Rapid Transit Deployment: Case Studies of Addis Ababa and Lagos
24. Planning-Oriented Operational Co-Optimization of Photovoltaic-Assisted Bus Charging and Swapping Systems
23. Joint Energy-Efficient Optimization of Electric Bus Scheduling and Charging Management
22. Charging Station Location Problem for Electric Buses with Opportunistic Fast Charging
21. Data-Driven Bi-Level Optimization for Multimodal Transit Networks: Integrating Trunk-Feeder Strategies and Mobile Signaling Data
20. Learning-Based Intelligent and Adaptive Dispatching for MaaS-Enabled Intermodal Rural Transit
19. When Should Service Zones Change? An Event-Triggered Adaptive Zoning Framework for Demand-Responsive Transit
18. Mobility Energy Productivity (MEP) Assessment of Rural Demand-Responsive Transit (DRT) Services
17. Modeling the Power Consumption of Electric Buses in Kigali, Rwanda
16. Bus-Priority Charging Coordination in Open Electric-Bus Depot Using a Graph Neural Network Matching Policy
15. Beyond Spatial Proximity: Classifying Taxi–DRT Market Overlap Using Observed Trip Data
14. Joint Design of Electric Bus Network and Charging Infrastructure Considering Wireless Charging atop a Hybrid Road-Network Structure
13. The Effect Of Central Charging Depot Location On Electric Bus Fleet Size Requirements For Fixed Routes
12. Operational and Energy Responses of Battery Electric Buses to Rainfall: A Vehicle-Specific Pattern Analysis

11. Mitigating Bus Bunching with Reinforcement Learning Enhanced by Semantic Stop Embedding
10. Evaluating Transit Bus Electrification through MOVES, GREET, and GIS: Emissions Reduction and Operational Readiness
9. Dynamic Shortest Path Accelerated Reinforcement Learning for Resilience Oriented Recovery Sequencing of Multimodal Transit Networks
8. Optimizing Passenger Flow Redistribution in Cascading Failures of Multimodal Transit Networks through Heterogeneity-Aware Physics-Informed Deep Learning
7. Joint Planning and Scheduling of Modular Vehicles for Passenger–Freight Integration
6. Transit Fleet Decarbonization Tool: Well-To-Wheel Ghg Accounting, Scenario Analysis, And Mode-Shift Assessment
5. From Nominal to Effective Shared-Capacity: Passenger-Preserving Transit Capacity-Sharing under Stochastic Overflow Risk
4. Transit-MP: Transit-Prioritized Max-Pressure Control in Sparse Connected Vehicle Environments
3. Mixed Diesel and Electric Bus Fleet Scheduling Problem for Pre-Typhoon Evacuation under Energy Uncertainty
2. Agentic Aggregation for Electric Bus Depot Coordination: Operational Trade-Offs, Contextual Pricing, and Deployment Implications